

*****REDHAT ENTERPRISE LINUX*****

*****CLASS 1 AWS ACCOUNT*****

AWS account creation login process etc

Save & visit once.

<https://www.awseducate.com/>

<https://aws.amazon.com/getting-started/>

<https://amazon.qwiklabs.com/>

Architecture of Linux <https://www.javatpoint.com/architecture-of-linux>

Linux File System Hierarchy <https://www.geeksforgeeks.org/linux-file-hierarchy-structure/>

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*****CLASS 2 BASIC COMMANDS*****

Command	Description
clear ↵	Clears the screen
date ↵	Displays the current time & date with time zone
cal ↵	Displays calendar on screen
bc ↵	Opens calculator (to exit press ctrl + c)
whoami ↵	Displays current username
w ↵	Displays how many users are logged in
hostname ↵	Displays hostname
hostname -i ↵ ifconfig ↵ ip addr ↵	Displays ip addresses Displays network interfaces and ip addresses of that interfaces
su USERNAME ↵	Switches to specified user
sudo -i ↵ sudo su ↵	Switches to root user
pwd ↵	Displays present working directory
ls ↵ ll ↵ ls -a ↵ ls -li ↵ ls -ltrZ ↵	Displays list of files and folders normal list Displays list of files and folders long list Displays list of files and folders with hidden items Displays inodes Displays list of files and folders with context value
cd SRC DEST ↵	Changes directory
cd ↵ cd ~ ↵	Changes directory to current user home directory
cd - ↵	Switches directory between last two opened directories
cd .. ↵ cd ../.. ↵ cd ../../.. ↵	Changes directory to one directory back Changes directory to two directories back Changes directory to three directories back
history ↵	Displays list of commands executed
exit ↵	Exits from current user or terminal if all users get logged out
touch FILENAME ↵ touch FILENAME1 FILENAME2 ↵ touch FILENAME{1..20}.ext ↵	Creates an empty file Creates 2 empty files at same time Creates multiple empty files at same time
cat FILENAME ↵	Displays content of specified file

cat > FILENAME ↵ CONTENT Ctrl+D	Creates file if it doesn't exist else overrides content of existing file
cat >> FILENAME ↵ CONTENT Ctrl+D	Creates file if it doesn't exist else appends content to existing file
echo "CONTENT" > FILENAME ↵	Creates file if it doesn't exist else overrides content of existing file
echo "CONTENT" >> FILENAME ↵	Creates file if it doesn't exist else appends content to existing file
mkdir DIRNAME ↵ mkdir DIRNAME1 DIRNAME2 ↵ mkdir DIRNAME{1..20} ↵	Creates a directory Creates 2 directories at same time Creates multiple directories at same time
rmdir DIRNAME ↵ rmdir DIRNAME1 DIRNAME2 ↵	Deletes a directory if it is empty else not Deletes 2 directories at same time if they are empty else not
rm -rf DIRNAME ↵ rm -rf DIRNAME1 DIRNAME2 ↵ rm -rf DIRNAME FILENAME ↵ rm -rf * ↵ rm -rf .* ↵	Deletes a directory with all contents (recursively) Deletes 2 directories with all contents at same time (recursively) Deletes directory & file specified (recursively) Deletes everything inside current directory except hidden files Deletes everything inside current directory with hidden files
mkdir .DIRNAME ↵ touch .FILENAME ↵	Creates a hidden folder Creates a hidden file
cp SRC DEST ↵ cp -r SRC DEST ↵	Copies file from SRC to DEST Copies directory with all its content (recursively)
mv SRC DEST ↵	Moves directory/file if SRC & DEST are different Renames directory/file if SRC & DEST are same
head FILENAME ↵	Displays top 10 lines of the file
tail FILENAME ↵	Displays bottom 10 lines of the file
more FILENAME ↵	Displays one screen at a time in case the file is large
less FILENAME ↵	Like more command but has features like scroll back & forth
file FILENAME ↵ file DIRNAME ↵	Displays what kind of file is specified file Displays whether it is directory or not
COMMAND --help ↵ man COMMAND ↵	Displays how to use a command
whatis COMMAND	Displays short description of the command specified
whereis COMMAND which COMMAND	Displays where the command is in bash
grep PATTERN FILENAME ↵ cat FILENAME grep PATTERN ↵ grep -i PATTERN FILENAME ↵ grep -ri PATTERN FILENAME ↵ grep -n -ri PATTERN FILENAME ↵	Find patterns in a file displays row wise Grep with cat command Displays without case sensitivity Displays recursively without case sensitivity Displays recursively line no's without case sensitivity
cat FILENAME awk -F ":" '{print \$1}' ↵	Works as grep to find text in a file but displays specific column wise output can integrate with grep also (: specifies what separates column)
cat FILENAME grep PATTERN awk -F ":" '{print \$1}' ↵	Same as above with grep integration
sed -i -e 's/PATTERN/REPLACE/g' FILENAME ↵	Finds pattern and replaces with the text specified in REPLACE area.
ex FILENAME <<eof ↵ LINENUMBER insert/delete ↵ CONTENT ↵ . ↵ xit ↵ eof ↵	Appends content at specified line number Deletes content at specified line number
export VARNAME="CONTENT" ↵ echo "VARNAME='CONTENT'" >> ~.bashrc ↵	To create temporary environment variables To create permanent environment variables

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echo "export VARNAME" >> ~/.bashrc ↵	
alias ALIASNAME='COMMAND' ↵	Creates temporary alias (gets cleared when logged out)
echo "alias ALIASNAME='COMMAND'" >> ~/.bashrc ↵	Creates permanent alias for user specified

*****CLASS 3 VI EDITOR*****

Command	Description
vi FILENAME ↵	Open or edit a file
i	Switch to Insert mode can use Insert Key also
Esc key	Switch to Command mode
Esc :w	Save and continue editing
Esc :x Esc :wq ZZ	Save and quit/exit vi
:q!	Quit vi and do not save changes
:q	Quit vi when no changes made
yy	Yank (copy) a line of text
p	Paste a line of yanked text below the current line
o	Open a new line under the current line
O	Open a new line above the current line
A	Append to the end of the line
a	Append after the cursor's current position
I	Insert text at the beginning of the current line
b	Go to the beginning of the word
e	Go to the end of the word
x	Delete a single character
dd	Delete an entire line
Xdd	Delete X number of lines
Xyy	Yank X number of lines
G	Go to the last line in a file
XG	Go to line X in a file
gg	Go to the first line in a file
:num	Display the current line's line number
:se nu	Display the line numbers
:NUMBER	Go to specified number, example :10
h	Move left one character
j	Move down one line
k	Move up one line
l	Move right one character

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*****CLASS 4 FILES & PERMISSIONS*****

Example file details below from ll command

```
-rw-r--r-- 1 root  root   55 Jul 30 14:29 text.txt
```

*****FILE TYPES*****

Symbol	Description
-	Regular or ordinary file
d	Directory / Folder
l	Link files (Symlinks)
b	Block special file
p	Named pipe file
c	Character special file
s	Socket file

*****FILE DETAILS*****

```
-rw-r--r-- 1 root  root   55 Jul 30 14:29 text.txt
```

Symbol	Description
-	File type
rw-r--r--	File permissions
1	No of links
root	User/Owner
root	Group
55	File size
Jul 30 14:29	Date & time at which the file is modified last
text.txt	File name

FILE PERMISSIONS			FILE PERMISSIONS ORDER			FILE PERMISSIONS ASSIGNMENT	
Type	Symbol	Value	USER(u)	GROUP(g)	OTHERS(o)	Symbol	Description
Read	r	4	r w -	r - -	r - -	+	Add
Write	w	2	4+2+0	4+0+0	4+0+0	-	Remove
Execute	x	1	6	4	4	=	OverRide

Command	Description
chmod PERMISSIONS FILENAME/DIRNAME ↵	Change permissions to file/directory
chmod -R PERMISSIONS DIRNAME ↵	Change permissions recursively to directory
chmod u+r FILENAME/DIRNAME ↵	Append permissions symbolic way to file/directory
chmod u-r FILENAME/DIRNAME ↵	Remove permissions symbolic way to file/directory
chmod ugo=rwx FILENAME/DIRNAME ↵	Override permissions symbolic way to file/directory

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chown USERNAME FILENAME/DIRNAME ↵	Change owner to a file/directory
chown :GROUP FILENAME/DIRNAME ↵	Change group to a file/directory
chown USERNAME:GROUP FILENAME/DIRNAME ↵	Change owner & group to a file/directory
chown -R USERNAME:GROUP FILENAME/DIRNAME ↵	Change owner & group to a file/directory recursively

*****CLASS 5 FIND, LINKS & UMASK*****

Command	Description
find PATH -name FILENAME ↵	Find files / directories based on name
find PATH -iname FILENAME ↵	Find files / directories based on name without case sensitivity
find PATH -type d -iname DIRNAME ↵	Find directories only
find PATH -type f -iname FILENAME ↵	Find files only
find PATH -type f -iname *.EXT ↵	Find files only with specified extensions
find PATH -type f -perm 644 ↵	Find files only with permissions 644
find PATH -type d -perm 644 -exec chmod 777 {} \; ↵	Find files with permissions 644 and change permissions to 777
find PATH -type f -perm 644 -exec rm -rf {} \; ↵	Find files with permissions 644 and delete those files
find PATH -mtime NO-OF-DAYS ↵	Find files based on modified time
ls -li ↵	Find inode no of file or directory
df -ih ↵	Find inodes no of disk
ln -s SRC SOFTLINKNAME ↵	Creates a softlink for specified source file
unlink SOFTLINKNAME ↵	Removes softlink
ln SRC HARDLINKNAME ↵	Creates a hardlink for specified source file (Hardlink has same inode number as source file)
unlink HARDLINKNAME ↵	Removes hardlink
umask ↵	To view default permissions of files/directories
umask 0022 ↵	To change default permissions

*****CLASS 6 STIKY BIT & USERS*****

StickyBit: By enabling this only the owner of the file and root user will have access to the directoryname or file in linux there is a directory with default sticky bit enabled -> /tmp/ (common interview question) to check this ll -ld /tmp/

Command	Description
chmod 1777 DIRNAME ↵ chmod o+t DIRNAME ↵	To enable Sticky Bit
chmod 0777 DIRNAME ↵ chmod o-t DIRNAME ↵	To disable Sticky Bit
ll -ld DIRNAME	To Check Sticky Bit

When a user is created a group with same username is created and the user is by default added to that group. A user can be a part of multiple groups (primary group & secondary group).

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There are 3 types of users in Linux.

Sl No	User Type	User ID
1	root	0
2	system	1 - 999
3	normal	1000-65000

File Name/Dir Name	Description (What it contains)
/etc/passwd	User Related Info (This file gets modified when a new user is created)
/etc/shadow	User Password Info (This file gets modified when a new user is created)
/etc/group	User Group Info (This file gets modified when a new user is created)
/etc/gshadow	User Group Info Pwd (This file gets modified when a new user is created)
/etc/default/useradd /etc/login.defs	Useradd Configuration
/etc/skel	.bashrc .bash_logout .bash_profile (If a new user is created these hidden files are copied to the /home/username from /etc/skel)

cat /etc/passwd **root:x:0:0:root:/root:/bin/bash**

Symbol	Description
root	Username
x	Password
0	User ID
0	Group ID
root	Comment
/root	User Home Directory
/bin/bash	Shell

Commands	Description
useradd USERNAME	Creates User
passwd USERNAME	Sets Password for the User
id USERNAME	Displays User ID & Group IDs
useradd -u 1011 USERNAME	To customize User ID & Group ID
useradd -g 1011 USERNAME	To customize Group ID if group already exists
useradd -c "FULLNAME" USERNAME	To customize comment
useradd -d /opt/USER USERNAME	To customize home directory
useradd -s /usr/bin/zsh USERNAME	To customize shell
usermod -u 1011 USERNAME	To modify User ID
usermod -g 1011 USERNAME	To modify Group ID
usermod -G 1000 USERNAME	To add secondary group
usermod -c "test" USERNAME	To modify comment
usermod -d /opt/USER USERNAME	To modify home directory
usermod -s /sbin/nologin USERNAME	To modify shell

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userdel USERNAME	To delete user only
userdel -r -f USERNAME	To delete user along with home directory

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*****CLASS 7 PASSWORDS & GROUPS*****

cat /etc/shadow

root:!:19625:0:99999:7:::

Symbol	Description
root	Username
*	Password
19625	Last password changed
0	Min no of days required to change password
99999	Max no of days required to change password
7	Warning for password change
	Inactive
	Expiry

To change password related parameters

Password Commands	Description
chage USERNAME	To change everything
chage -l USERNAME	To check all
chage -E YYYY-MM-DD username	To change only password expiry date
chage --help	Try all parameters

cat /etc/group

root:x:0:

Symbol	Description
root	Group name
x	Password
0	Group ID
	List of users in the group

Command	Description
groupadd GROUPNAME	To ceate a group
groupadd -g GID GROUPNAME	Customize group id
groupadd -r GROUPNAME	To create system group
groupmod -g GID USERNAME	To modify primary group
groupmod -g GID GROUPNAME	To modify group id
groupmod -n OLDGROUPNAME NEWGROUPNAME	To modify group name

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gpasswd	Group administration
gpasswd -a USERNAME GROUPNAME	Add a user to group
gpasswd -M USER1, USER2, .. USERN GROUPNAME	Add multiple users to a group (overwrites existing users)
gpasswd GROUPNAME	Set password to a group
gpasswd -r GROUPNAME	Remove password from a group
gpasswd -A USERNAME GROUPNAME	To set admin to group

Tasks	Description
1 (Task No 38)	Add multiple users to group without overwriting existing users
2 (Task No 39)	Add multiple admins to single group
3	Delete admin user from the group
4	Remove admin privileges from the group

*****CLASS 8 SUDOERS & CRONJOB*****

useradd hi To create user normal hi
su hi To swith user to hi
cd /etc/ Change dir to /etc/
touch file.txt Try to create file.txt (we cannot create file)

to achieve this, we need root permissions, so we need to use sudo command before command

sudo touch file.txt

Now will have root privilege to create file even after this we will not be able to create a file because user hi is not in sudoers list, so we need to add user hi to sudoers list located at /etc/sudoers we need to make entries for hi user. directly editing /etc/sudoers file is not recommended as it may corrupt the file to edit that file, we have single command called visudo.

visudo

We need to make entries below ## Allow root to run any commands anywhere.

root ALL=(ALL) ALL

hi ALL=(ALL) ALL (this will enable hi user to have root access but we need to use sudo before command and it will ask for password)

hi ALL=(ALL) NOPASSWD: ALL (no password prompt)

To give access to only specified commands

which mkdir to check bin path of the command

hi ALL=(ALL) NOPASSWD: /usr/bin/mkdir, /usr/bin/touch, /usr/bin/ls, /usr/bin/cat

Cronjob: The cron command-line utility is a job scheduler on Unix-like operating systems. Users who set up and maintain software environments use cron to schedule jobs, also known as cron jobs, to run periodically at fixed times, dates, or intervals.

1. cronjob To schedule a task on linux os
2. **service crond status** must be in active status to use cron job

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3. cron expression min, hours, day of month, month, day of week
4. **crontab -e** To edit the crontab
5. **crontab -l** To list the crontabs
6. **/var/log/cron** log file (to check cronjob is running or not)
7. crontabs are user specific & convert to your local time from server time

Cron Expression	Description
* * * * * COMMAND	Cron exp for running every min, hour, day, mon, week
30 22 * * * COMMAND	Cron exp for running every day 10 30 pm
28 23 24 Aug * COMMAND	Cron exp for running every 24 of Aug 11:28 PM
* / 5 * * * * COMMAND	Cron exp for running every 5 mins
* * / 2 * * * COMMAND	Cron exp for running every 2 hrs
30 18 22 Nov Wed COMMAND	Cron exp for running on Nov 22 Wed 6:30 PM
* * * Aug,Sep,Oct,Nov * COMMAND	Cron exp for running on Aug Sep Oct Nov
* * * Aug-Dec * COMMAND	Cron exp for running on Range Aug to Dec
00 05 * * Mon-Fri	Daily 10 30 AM IST Monday - Friday (First Convert - 10:30 AM IST to UTC)

@min @hourly @month @yearly

Command	Description
crontab -l	To check existing crontabs
crontab -e * * * * * df -h > /tmp/disk.txt	To create new crontab
service crond stop	to stop crontab Try stopping crond service and see the result it will not execute
crontab -u USERNAME -l	To view cronjob of diff user from root
/var/spool/cron/	Path where all cronjobs are stored for each user
crontab -u USERNAME -e	To edit cronjob of diff user from root
crontab -u USERNAME -e	To remove crontab remove that line that deals with the job or comment with ##
crontab -r	To remove crontab completely
/etc/cron.deny	To deny user just enter username in each line
/etc/cron.allow	To allow user just enter username in each line (this file doesn't exist we need to create it)

See below for examples.

<https://crontab.guru>

<https://tecadmin.net/crontab-in-linux-with-20-examples-of-cron-schedule/>

Tasks	Description
1	To run task for every 30 secs
2	Schedule a cron for a system that got shutdown before making an entry and it should run after starting.

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1. rpm - RedHat Package Manager (Cannot Resolve Dependencies)

Command	Description
yum install wget -y	To wget package
wget URL	To download data/packages etc from internet
rpm -ivh PACKAGENAME	To install package
rpm -qpr PACKAGENAME	To list compatibilities of package & its dependencies
rpm -ivh PACKAGENAME -nodeps	To install package without dependencies
rpm -q PACKAGENAME	To check package
rpm -ql PACKAGENAME	Query list package files list
rpm -qa -last	Recently installed packages
rpm -Uvh PACKAGENAME	To update specific package
rpm -eVV PACKAGENAME	To remove specific package
rpm -qf PACKAGENAME	To query file
rpm -qi PACKAGENAME	To get rpm package info
rpm -qip PACKAGENAME	To get rpm package info before installation
rpm -Vp PACKAGENAME	To verify package name
rpm -qa PACKAGENAME	To find package installed or not
rpm -qa --last grep -i PACKAGENAME	To find package when installed or when updated

2. yum Yellowdog Updater Modified (Can Resolve Dependencies)

Command	Description
/etc/yum.repos.d/	all repos urls are stored in this directory
yum install PACKAGENAME -y	To install package
yum remove PACKAGENAME -y	To remove package
yum update PACKAGENAME -y	To update package
yum update -y	To update all package
yum list PACKAGENAME	To list package
yum list installed	To list installed packages
yum list	To list all packages
yum list installed grep PACKAGENAME	To list installed package
yum search PACKAGENAME	To search for package
yum provides COMMAND	To check which package, provides COMMAND
yum info PACKAGENAME	To check package info
yum check update	To check for update
yum history	To check yum history
yum undo PACKAGENAME-VERSION	To downgrade PACKAGENAME
yum clean all	To clean cache

Command	Description
yum install zip -y	Install zip
zip FILENAME.zip FILES / DIRECTORIES	create zip file
zipinfo FILENAME.zip	View content of zip
zip ALL.zip file1.txt file2.txt	To zip more than 1 file
zip ALL.zip *	To zip all files and folders
zip -u ALL.zip file10.txt	To update existing zip file
zip -d ALL.zip cron.txt	To delete existing files in zip file
zip -r TEST.zip test/	To zip a directory recursively
zip -m ALL1.zip *	To zip files and delete them
zip /home/ec2-user/USER.zip users.txt	To zip files in a diff location
zip -rm DIR.zip * -x users.txt	Exclude file from zipping
zip -r DIR.zip * -x DIRNAME*	Exclude dir from zipping
zip --help	Explore
unzip DIR.zip	To extract
unzip -l DIR.zip	To view contents
unzip -Z DIR.zip like zipinfo	To view detailed contents
unzip DIR.zip -d /tmp/	To unzip at diff location
unzip DIR.zip users.txt	To unzip selected file
unzip DIR.zip users.txt -d /mnt/	To unzip selected file diff loc
unzip DIR.zip -x users.txt	To exclude file in unzip
unzip DIR.zip -x DIRNAME*	To exclude dir in unzip
unzip -q DIR.zip	To unzip in suppress mode
tar -cvf DIR.tar cron.txt	To compress highly
tar -tvf DIR.tar	To view
tar -rvf DIR.tar cron1.txt	To add file to existing tar
tar -xvf DIR.tar	To extract
tar -cvzf DIR.tar.gz *	To zip in tar.gz
du -sch DIR.tar.gz	To check size
tar -xvzf DIR.tar.gz	To unzip tar.gz
tar -cvzf DIR.tar.tgz *	To compress more

Task 1. Zip files and folders at the same time.

Bootimg - Process of loading operating system from the hard disk into the ram.

<https://www.thegeekstuff.com/2011/02/linux-boot-process/>

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To connect a server, we have 3 ways:

1. Using Private Key (Private Key Based Authentication)

This is the default authentication that aws provides, while launching an instance we will create a keypair of our choice (pem/ppk) it will download that keypair on to our local machine the keypair that gets downloaded is Private Key, using this keypair we will login to the server using Putty or MobaXTerm, this is Private Key Based Authentication.

2. Using Public Key (Public Key Based Authentication)

After login with above method (Private Key Based Authentication) we will copy the public key (~/.ssh/authorized_keys) of the user and append the key to the same location of another server or local machine (~/.ssh/authorized_keys) after this we need to make necessary changes to the ssh configuration file (/etc/ssh/sshd_config) like **PubKeyAuthentication yes** and restart sshd service with **systemctl restart sshd**. In some cases, we may also need the private key to be copied to the respective server or local machines to avoid errors. If there is no public-key, then we can directly create that file with contents of another server or local machine.

3. Using Password (Password Based Authentication)

After login with above method (Private Key Based Authentication) we will set password for the user and make necessary changes to the ssh configuration file (/etc/ssh/sshd_config) like **PasswordAuthentication yes** and restart sshd service with **systemctl restart sshd**.

Tasks to do:

1. Launch 2 ec2 instances
2. Use MobaXTerm to login
3. Set hostname for both servers
4. Create users in both servers
5. Generate ssh keys in server A and copy public key to server B
6. Generate ssh keys in server B and copy public key to server A

Password-less Authentication

Server - A -> user1 -----> Server - B -> user2

- > In Server - A Set Hostname `hostnamectl set-hostname server-a && exec bash`
- > In Server - A `useradd user1`
- > Set password `passwd user1`
- > Switch to user user1 `su user1`
- > Generate ssh keys using `ssh-keygen`
- > It creates .ssh directory with perm 700 and inside of .ssh it creates `id_rsa` and `id_rsa.pub`
- > Rename ~/.ssh/id_rsa to authorized_keys `mv ~/.ssh/id_rsa.pub ~/.ssh/authorized_keys`
- > Change permission to 600 must `chmod 600 ~/.ssh/authorized_keys`
- > Need to copy public key from server -A user1 to Server -B user2 -> `ssh-copy-id user2@PUB-IP-SRV-B`
- > In server B it will create .ssh directory with perm 700 under home directory of user2 and inside of .ssh directory it will create one file called `authorized_keys`
- > In Server - A `ssh user2@PUB-IP-SRV-B`

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Server - B -> user2 -----> Server - A -> user1

- > In Server - B Set Hostname `hostnamectl set-hostname server-b && exec bash`
- > In Server - B `useradd user2`
- > Set password `passwd user2`
- > Switch to user user2 `su user2`
- > Generate ssh keys using `ssh-keygen`
- > It creates `.ssh` directory with perm 700 and inside of `.ssh` it creates `id_rsa` and `id_rsa.pub`
- > Rename `~/.ssh/id_rsa` to `authorized_keys` `mv ~/.ssh/id_rsa.pub ~/.ssh/authorized_keys`
- > Change permission to 600 must `chmod 600 ~/.ssh/authorized_keys`
- > Need to copy public key from Server-B user2 to Server-A user1 -> `ssh-copy-id user2@PUB-IP-SRV-A`
- > In server A it will create `.ssh` directory with perm 700 under home directory of user1 and inside of `.ssh` directory it will create one file called `authorized_keys`
- > In Server - B `ssh user2@PUB-IP-SRV-A`

Change to password-based authentication.

`vi /etc/ssh/sshd_config` -> `PasswordAuthentication yes && service sshd restart`

Change to password-less authentication.

`vi /etc/ssh/sshd_config` -> `PubKeyAuthentication yes && service sshd restart`

To check connectivity between to servers we need telnet (to install `yum install telnet -y`)

`telnet PRI-IP-SRV-B` - In Server A `telnet PRI-IP-SRV-A` - In Server B

Command	Description
<code>scp LOCALPATH USERNAME@REMOTE-IP:REMOTEPath</code>	Secure Copy (scp) from local machine to remote server
<code>scp USERNAME1@REMOTE1-IP:REMOTEPath1 USERNAME2@REMOTE2-IP:REMOTEPath2</code>	Secure Copy (scp) from remote server to remote server
<code>scp USERNAME@REMOTE-IP:REMOTEPath LOCALPATH</code>	Secure Copy (scp) from remote server to local machine
<code>rsync -r DIRNAME1 DIRNAME2</code>	rsync local recursively copies files
<code>rsync -r -a DIRNAME1 DIRNAME2</code>	rsync local recursively copies files with archive mode
<code>rsync -r LOCALPATH USERNAME@ REMOTE-IP:REMOTEPath</code>	rsync from local machine to remote server
<code>rsync USERNAME@REMOTE-IP:REMOTEPath LOCALPATH</code>	rsync from remote server to local machine
<code>scp -help</code> <code>rsync --help</code>	Explore scp Explore rsync

SCP (Secure Copy)

SCP is a Secure Copy Protocol. It is a command line utility that allows the user to securely copy files and directories between two locations usually between unix or linux systems.

RSYNC

Rsync, which stands for remote sync, is a remote and local file synchronization tool. The syntax for rsync operates like other tools, such as ssh, scp, and cp.

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1. Monitoring Tools:

- a. **top**: The top command is used to display all the running and active real-time processes in an ordered list and updates it regularly. It displays CPU usage, Memory usage, Swap Memory, Cache Size, Buffer Size, Process PID, User, Commands, and much more.
- b. **vmstat**: Linux VmStat command is used to display statistics of virtual memory, kernel threads, disks, system processes, I/O blocks, interrupts, CPU activity, and much more. To get vmstat command need to install a package called sysstat
- c. **lsof**: to display a list of all the open files and the processes. The open files included are disk files, network sockets, pipes, devices, and processes.
- d. **netstat**: Netstat - derived from the words network and statistics - is a command-line utility used by system administrators for analyzing network statistics. It displays a whole manner of statistics such as open ports and corresponding addresses on the host system, routing table, and masquerade connections.
- e. **iostat**: iostat is a simple tool that will collect and show system input and output storage device statistics. This tool is often used to trace storage device performance issues including devices, local disks, remote disks such as NFS.
- f. **route**: Using route to Display Routing Information

When route is used with no parameters, the route command prints the routing table to stdout:

Command	Description
Uptime	When the system in powered on
top	CPU & Memory usage (load average last 5 min 10 mins 15 mins) Swap Memory used for storing sleep processes
free -k free -m free -g	Memory size & Swap size in kbs Memory size & Swap size in mbs Memory size & Swap size in gbs
cat /proc/cpuinfo	Complete system inf
cat /proc/meminfo	Complete mem info
vmstat iostat	Statistics of vm (to install yum install sysstat -y) SAR Utility
lsof	List all open files etc (to install yum install lsof -y)
netstat	Network statistics (to install yum install net-tools)
netstat -tnlap	To check tcp protocol
netstat -unlap	To check udp protocol
iostat	To chek input output stat
route	To display routing info
df -h	Displays disk & mnt info ROM
du -bsh *	Disk usage in bytes

2. Process Management:

- A Linux process is a program running in the Linux system. a Linux process is also called daemon.
- In Linux, every running process or daemon is given an identity number called PID (Process ID).
- The process id is unique we can terminate unused program in the system by stopping its process id.

- In order to manage Linux processes, we need to identify some process information such as who's responsible for the process, which terminal the process is running from and what command used to run the process.

There are generally three types of processes that run on Linux.

- Interactive Processes
- System Process or Daemon
- Automatic or batch

a. Interactive Processes:

Interactive processes are those processes that are invoked by a user and can interact with the user. VI is an example of an interactive process. Interactive processes can be classified into foreground and background processes. The foreground process is the process that you are currently interacting with and is using the terminal as its stdin (standard input) and stdout (standard output). A background process is not interacting with the user and can be in one of two states - paused or running.

b. System Process or Daemon:

The second general type of process that runs on Linux is a System Process or Daemon (day-mon). Daemon is the term used to refer to processes that are running on the computer and provide services but do not interact with the console. Most server software is implemented as a daemon. Apache, Samba, and jobs are all examples of daemons.

c. Automatic Processes:

Automatic processes are not connected to a terminal. Rather, these are tasks that can be queued into a spooler area, where they wait to be executed on a FIFO (first-in, first-out) basis. Such tasks can be executed using one of two criteria:

- At certain date and time: done using the "at" command
- At times when the total system load is low enough to accept extra jobs: done using the Cron command.

d. Parent and Child Process:

- The Process which starts or creates another process is called parent process and the one which got created is known as child process.
- Every process will be having a parent process except init process.
- The init process is the parent of all the process in the system. It is the first process which gets started by the kernel at the time of booting.
- The PID of init will be 1.
- Only after init process gets started the remaining process are called by it, and hence it is responsible for all the remaining processes in the system.

Command	Description
ps	The ps command gives the running process of the present terminal and present command
ps -a	To see total number of processes running in the system
ps -u USERNAME	To see the processes running by the logged in user (ex root)
ps -u	If no name is given it will show the processes of the logged in user
ps -x	To see which process are attached with some terminals (tty = TeleTypewriter) and which are not
ps -G USERNAME pgrep -G musab	To see which process are running by a particular group
ps -aux	To see the offline process of the system, already executed

3. Signals in Linux:

Signals can be generated by the process itself, or they can be sent from one process to another. A variety of signals can be generated or delivered, and they have many uses for programmers. (To see a complete list of signals in the Linux® environment, uses the command kill -l.)

Command	Description
kill -l	There are, total 64 signals in Linux, the list of all the signal
kill -9 11591	For killing the process
kill -15 PID	Terminating the process
kill -1 PID	For reloading the process
kill -20 PID	For stopping the process

Example:

To stop the process using a signal no. 20

User console: # su - ec2-user \$ cat > hello

Root console: # ps -u ram # kill -20 12041

User console: \$ cat > hello

[1] + Stopped cat > hello

\$ fg 1 Restart the process continue working cat > hello

=====

*****CLASS 13 LVM*****

1. Logical Volume Manager (LVM):

The Linux Logical Volume Manager (LVM) is a mechanism to virtualize the disks. It can create "virtual" disk partitions out of one or more physical hard drives, allowing you to grow, shrink, or move those partitions from drive to drive as your needs change. It also allows you to create larger partitions than you could achieve with a single drive. (to install lvm yum install lvm2* -y)

Command	Description
pvs	Displays all the physical volumes
pvdisplay	Displays detailed information on physical volumes
pvcreate	Create a new physical volume
pvmove	Moves/migrates data from one physical volume to another
pvscan	List all physical volumes
pvremove	Deletes a physical volume
vgs	Displays all volume groups in the system
vgdisplay	Displays detailed information on volume groups
vgcreate	Create a new volume group
vgextend	Add a new physical disk to a volume group
vgscan	Search for all volume groups
vgreduce	Reduces a volume group by removing a PV from it

vgremove	Removes /Deletes a volume group
lvs	Displays all the logical volumes in the system
lvdisplay	Displays detailed information on logical volumes
lvcreate	Creates a new logical volume
lvscan	Scan (all disks) for Logical Volumes
lvextend	Extends a logical volume
lvresize	Resizes a logical volume
lvreduce	Reduces a logical volume
lvremove	Removes /Deletes a logical volume

Command	Description
fdisk -l	Lists the path of raw disk
lsblk	To list disks & partitions
pvs / pvdisplay	To show pvs
pvcreate /dev/xvdf	To create pv
pvs	To check pvs
pvcreate /dev/xvdb /dev/xvdc /dev/xvdd	To create pv on multiple disks
vgcreate myvg /dev/xvdb	To create vg
vgreate myvg /dev/xvdb /dev/xvdc /dev/xvdd	To create vg on multiple disks
vgs	To check vgs
lvcreate --size 1G --name mylv1 myvg	To create lvm
lvcreate -L 5G -n mylv1 myvg	To create lvm
vgs	To check vgs
mkfs -t ext4 /dev/myvg/mylv1	To crate fs
mount /dev/myvg/mylv1 /mnt/	To mount lvm
unmount /mnt	To unmount
lvextend -L +500M /dev/myvg/mylv1	To extend lvm
lvreduce -L -500M /dev/myvg/mylv1	To reduce lvm
lvremove --force myvg	To remove lv
vgremove myvg	To remove vg
resize2fs /dev/myvg/mylv1	To resize fs in ext4
xfs_growfs /dev/myvg/mylv1	To resize fs in xfs

1. PV Create:

fdisk -l

pvcreate /dev/sda7 /dev/sda8 /dev/sda9

pvs or pvdisplay

2. VG Create:

```
# vgcreate myvg /dev/sda7 /dev/sda8 /dev/sda9
```

```
# vgs
```

```
# vgdisplay myvg
```

3. LV Create:

```
# lvcreate --size 100M --name mylv myvg
```

```
# lvs
```

```
# lvdisplay ktvvg      (to display the LVs of a particular volume group)
```

4. Create a file System:

```
# mkfs.ext4 /dev/ktvg/ktlv
```

5. Mounting:

```
# df -h
```

```
# mount /dev/ktvg/ktlv /ktdir
```

```
# df -h
```

```
# vim /etc/fstab
```

```
=> /dev/ktvg/ktlv    /ktdir ext4    defaults    0 0
```

```
# lvextend -L +200M /dev/ktvg/ktlv
```

```
# df -h
```

```
# resize2fs /dev/ktvg/ktlv
```

```
# df -h
```

```
# lvreduce -L -200M /dev/ktvg/ktlv
```

Purging:

```
# lvs
```

```
# lvremove /dev/myvg/mylv1
```

```
# lvremove /dev/myvg/mylv2
```

```
# vgs
```

```
# vgremove myvg
```

```
# pvs
```

```
# pvremove /dev/xvdb
```

```
# lsblk
```

```
=> Detach the disk from instance in AWS or VM.
```

=====

*****REDHAT ENTERPRISE LINUX*****

*****CLASS 14 PATCHING*****

Patching:

The below pre-patch tasks need to be completed 1day before the patching date:

- Confirm there is sufficient free space on / "Root Filesystem" and /boot.
- Confirm Console connectivity.
- Record License Subscription status and confirm the server can download the required packages from RHN. (#subscription-manager list)
- Linux Team will reconfirm the presence of the representative from the Application/DBA Team
- Download the patches. (#yum update -downloadonly)

The below pre-patch tasks need to be done 10 minutes before the patching:

Collect current system state using the following commands.

(#uptime ; free ; df -H ; netstat -nr ; ps -ef ; uname -a ; cat /etc/fstab ; mount)

Note: save the output of the previous commands in a text file using the name convention.

- Disable Nagios notifications for the respective server being patched. (disable is there any auto start trigger in Opswise).
- Request for a confirmation from the Application representative to start the patching because they must stop their APP/DB before patching. And in case of Infra server, Linux team is responsible to stop the service (Like Opswise).

The below activities should be followed, after patching the server.

- Application representative that the server is up to start their APP. And in case of Infra server, Linux team is responsible to start the service.
- Request a confirmation from the Application representative that all the related services and application are up. (This update needs to be updated on the Ticket)
- Check & confirm if all the services listed in the details document are currently running.
- Copy the content of /var/log/yum.log and save it in a text file with the naming convention.

Collect current system state and yum history transaction id using the following commands save it in a text file with the naming convention of POST-SITE-QUATER-SERVERNAME

(# uptime ; free ; df -H ; netstat -nr ; ps -ef ; uname -a ; cat /etc/fstab ; mount ; sudo yum history)

=====

*****CLASS 15 FILE SYSTEMS*****

Class is about Task 45 - Add second volume to linux instance and mount it on /mnt/

Command	Description
fdisk -l	List disks
fdisk /dev/xvdb	Create partition (enter m follow procedure)
mkfs.ext4 /dev/xvdb mkfs -t ext4 /dev/xvdb	Create filesystem
mount /dev/xvdb /mnt/	To mount filesystem
etc/fstab	To make kernal level mounting mount upon reboot (edit accordingly)

=====